FIRST SEMESTER M.Sc. DEGREE EXAMINATION, DECEMBER 2019

(CUCSS)

Mathematics

MT 1C 05—DISCRETE MATHEMATICS

(2016 Admissions)

Time : Three Hours

Maximum : 36 Weightage

Part A (Short Answer Questions)

Answer **all** *questions.*

Each question has weightage 1.

1. Compute the number of edges in the complete bipartite graph $K_{m,n}$.
2. In any simple graph, prove that the sum of the degrees of the vertices of a graph G is twice the number of edges.
3. Prove that a graph with at least 3 vertices is 2-connected if and only if any two vertices of G lie on a common cycle.
4. Define cut vertex and cut edge. Explain with an example.
5. Let v be a cut point of a connected graph G. Then, prove that there exist points u and w distinct from v such that v is on every $u-w$ path.
6. Prove that graph $K_{3,3}$ is non-planar.
7. Explain differences between a partial order relation and an equivalence relation.
8. Let P be the set of all real valued functions defined on a non-empty set X and $\leq$ be defined on P by $f \leq g$ if $f(x) \leq g(x)$ for every $x \in X$. Under what conditions $(P, \leq)$ is a chain.
9. If P is a partial order on the set X, show that $P \setminus \{(x,x) : x \in X\}$ is a strict partial order.
10. Define a Boolean algebra.
11. Define a Boolean function on n variables. Give an example of a Boolean function of 4 variables.
12. If $L = \{e, f\}$, find L^R and L^2.
13. Define *dfa*.
14. If $\Sigma = \{0,1\}$, design an *nfa* to accept set of strings either ending with two consecutive ones or two consecutive zeros.

(14 × 1 = 14 weightage)

Turn over

Part B (Paragraph Type)

Answer any **seven** *questions from the following ten questions.*
Each question has weightage 2.

15. Let G be a self-complementary graph of order n. Then, prove that $n \equiv 0,1 \pmod 4$.

16. Prove that a tree of order n has $n-1$ edges, $n \in \mathbb{N}$.

17. With usual notation, prove that $k(G) \leq \lambda(G)$, in any simple graph G.

18. Prove that every maximal outer planar graph G with p points has $2p-3$ edges.

19. Prove that the Petersen graph is non-planar.

20. Define a chain in a poset. Prove that the intersection of two chains is a chain.

21. Let $(X, +, .,')$ be a Boolean algebra. Prove that $x + x.y = x$ for all $x, y \in X$.

22. Prepare the table for values of the function $f(x,y,z) = (x+y)z'$.

23. Find a *dfa* that accepts all strings on $\Sigma = (a,b)$ starting with the prefix *aa*.

24. Find a *dfa* for the language $L = \{ab^4\omega b^2 : \omega \in \{a,b\}\}^*$.

(7 × 2 = 14 weightage)

Part C (Essay Type)

Answer any **two** *questions from the following four questions.*
Each question has weightage 4.

25. Define line graph of a graph. If G_1 and G_2 are isomorphic simple graphs, then prove that $L(G_1)$ and $L(G_2)$ are isomorphic.

26. Prove that a non-trivial connected graph is Eulerian if and only if all its vertices are of even degree.

27. Prove that every Boolean algebra is isomorphic to a power set Boolean algebra.

28. Explain with an example how to convert an *nfa* into an equivalent *dfa* using transition graph.

(2 × 4 = 8 weightage)